

Exercise #1

Suppose that a routine is available that computes numerically the pseudoinverse of a matrix A , e.g., the `pinv` function in MATLAB, $A^\# = \text{pinv}(A)$. Given a $m \times n$, full rank matrix J , with $m < n$, and a $n \times n$, positive definite, symmetric weighting matrix W , prove formally that the weighted pseudoinverse $J_W^\#$ can be computed as $J_W^\# = W^{-1/2} \text{pinv}(JW^{-1/2})$. As a verification, provide a simple numerical example with $m = 2$, $n = 3$.

Since J is full rank $J^\# = J^T(JJ^T)^{-1}$, the matrix JW is still a $m \times n$ nonsingular matrix.

$$J_W^\# = W^{-1} J^T (J W^{-1} J^T)^{-1}$$

$$W^{-1/2} (J W^{-1/2})^\# = W^{-1/2} (J W^{-1/2})^T [J W^{-1/2} (J W^{-1/2})^T]^{-1} =$$

$$W^{-1/2} W^{-1/2} J^T [J W^{-1/2} W^{-1/2} J^T]^{-1} =$$

$$W^{-1} J^T (J W^{-1} J^T)^{-1} = J_W^\# \quad \blacksquare$$

Exercise #2

A single link mounted on a passive elastic support is moved on a horizontal plane by a torque τ applied by a motor to the revolute joint at its base, as sketched in Fig. 1. The generalized coordinates q_1 and q_2 are defined therein, together with the relevant dynamic parameters: mass m , distance $d > 0$ of the CoM from the joint, and barycentric inertia I_L of the link; stiffness $k > 0$ of the linear spring in the support. The spring is undeformed when $q_1 = 0$. Derive first the Lagrangian dynamic model of this simple robotic system. Next, address the following two dynamic problems.

- a) An input torque $\tau_0 > 0$ is applied at $t = 0$, with the system at $q(0) = \dot{q}(0) = 0$ (zero initial state). Determine if the spring will initially be compressed or extended and if the link will start moving clockwise or counterclockwise. Provide the expression of the initial accelerations $\ddot{q}_1(0)$ and $\ddot{q}_2(0)$ of the two coordinates.

b) Starting at $t = 0$ in a generic non-zero initial state $(q(0), \dot{q}(0))$, define a control law $\tau = \tau(q, \dot{q})$ such that $q_2(t)$ will exponentially converge to zero. At steady state, determine the residual dynamics of the other coordinate $q_1(t)$ and provide a physical interpretation of this result.

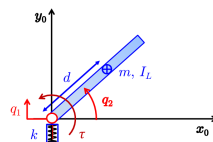
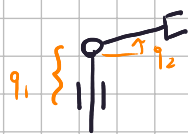


Figure 1: A single-link robotic system mounted on a flexible base.

$$q^T K q$$



$$P_c = \begin{Bmatrix} d c_2 \\ d s_2 + q_1 \end{Bmatrix} \Rightarrow \dot{P}_c = \begin{Bmatrix} -d s_2 \dot{q}_2 \\ d c_2 \dot{q}_2 + \dot{q}_1 \end{Bmatrix} \quad \dot{P}_c^T \dot{P}_c = d^2 \dot{q}_2^2 + \dot{q}_1^2 - 2 d c_2 \dot{q}_1 \dot{q}_2$$

$$T = \frac{1}{2} I_L \dot{q}_2^2 + \frac{1}{2} m (d^2 \dot{q}_2^2 + \dot{q}_1^2 - 2 d c_2 \dot{q}_1 \dot{q}_2) \Rightarrow$$

$$M_{11} = m \quad M_{22} = I_L + m d^2 \quad M_{12} = m d c_2 \Rightarrow M = \begin{pmatrix} e_1 & e_2 c_2 \\ e_2 c_2 & e_3 \end{pmatrix}$$

$$U = m g_0 (d s_2 + q_1) + \frac{1}{2} K q_1^2$$

$$\frac{\partial U}{\partial q_1} = m g_0 + K q_1 \quad \frac{\partial U}{\partial q_2} = m g_0 d c_2$$

$$C_1 = \frac{1}{2} \left(\frac{\partial M_{11}}{\partial q_1} + \left(\frac{\partial M_{11}}{\partial q_1} \right)^T - \frac{\partial M_{11}}{\partial q_1} \right) = \begin{pmatrix} 0 & 0 \\ 0 & -e_2 s_2 \end{pmatrix} \Rightarrow c_1 = -e_2 s_2 \dot{q}_2^2$$

$$C_2 = \frac{1}{2} \left(\frac{\partial M_{22}}{\partial q_2} + \left(\frac{\partial M_{22}}{\partial q_2} \right)^T - \frac{\partial M_{22}}{\partial q_2} \right) = 0 \quad \downarrow \quad M \ddot{q} + c + g = \begin{pmatrix} 0 \\ \tau \end{pmatrix}$$

$$\begin{cases} e_1 \ddot{q}_1 + e_2 c_2 \ddot{q}_2 + m g_0 + K q_1 - e_2 s_2 \dot{q}_2^2 = 0 \\ e_2 c_2 \ddot{q}_1 + e_3 \ddot{q}_2 + m g_0 d c_2 = \tau \end{cases}$$

a) At rest we have

$$M(0) \ddot{q} + g(0) = \begin{pmatrix} 0 \\ \tau_0 \end{pmatrix} \Rightarrow \begin{pmatrix} e_1 & e_2 \\ e_2 & e_3 \end{pmatrix} \ddot{q} + \begin{pmatrix} m g_0 \\ m g_0 d \end{pmatrix} = \begin{pmatrix} 0 \\ \tau_0 \end{pmatrix} \Rightarrow$$

$$\ddot{q} = M^{-1} \left(\begin{pmatrix} 0 \\ \tau_0 \end{pmatrix} - \begin{pmatrix} m g_0 \\ m g_0 d \end{pmatrix} \right) = \frac{1}{e_1 e_3 - e_2^2} \begin{pmatrix} e_3 & -e_2 \\ -e_2 & e_1 \end{pmatrix} \begin{pmatrix} -m g_0 \\ \tau_0 - m g_0 d \end{pmatrix} =$$

$$\frac{1}{e_1 e_3 - e_2^2} \begin{pmatrix} -m g_0 e_3 - e_2 \tau_0 + e_2 m g_0 d \\ m g_0 e_2 + e_1 \tau_0 - e_1 m g_0 d \end{pmatrix} \quad \text{notice how } e_1 e_3 - e_2^2 = I_L m > 0$$

$$\Rightarrow \begin{cases} \ddot{q}_1 = -m g_0 (I_L + m d^2) - m d \tau_0 + m^2 d g_0 \\ \ddot{q}_2 = m^2 g_0 d + m \tau_0 - m^2 g_0 \end{cases} \quad \tau_0 > m g_0 - m g_0 d$$

$$\ddot{q}_2 > 0 \Leftrightarrow \tau_0 > m g_0 (1 - d)$$

$$\ddot{q}_1 > 0 \Leftrightarrow \tau_0 < m g_0 - \frac{g_0}{d} (I_L + m d^2)$$

b) We Feedback linearize q_2 :

$$\ddot{q}_1 = \frac{1}{e_1} (e_2 s_2 \dot{q}_2 - m g_0 - K q_1 - e_2 c_2 \ddot{q}_2) \Rightarrow$$

$$\frac{e_2 c_2}{e_1} (e_2 s_2 \dot{q}_2 - m g_0 - K q_1 - e_2 c_2 \ddot{q}_2) + e_3 \ddot{q}_2 + m g_0 d c_2 = \tau \Rightarrow$$

$$\frac{e_2 c_2}{e_1} (e_2 s_2 \dot{q}_2 - m g_0 - K q_1) - \left(\frac{e_2^2 c_2^2}{e_1} + e_3 \right) \ddot{q}_2 + m g_0 d c_2 = \tau \Rightarrow$$

$$\tau = \frac{e_2 c_2}{e_1} (e_2 s_2 \dot{q}_2 - m g_0 - K q_1) + m g_0 d c_2 - \left(\frac{e_2^2 c_2^2}{e_1} + e_3 \right) a \Rightarrow$$

$$\ddot{q}_2 = a \quad \text{then} \quad a = -K_P q_2 - K_D \dot{q}_2.$$

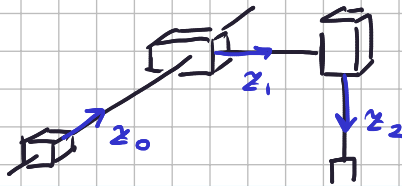
Exercise #3

Draw a 3-dof robot whose dynamic model is given by

$$M\ddot{q} + g = u, \quad M = \begin{pmatrix} m_1 + m_2 + m_3 & 0 & 0 \\ 0 & m_2 + m_3 & 0 \\ 0 & 0 & m_3 \end{pmatrix}, \quad g = \begin{pmatrix} 0 \\ 0 \\ -m_3 g_0 \end{pmatrix},$$

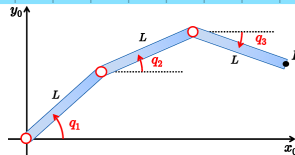
with masses $m_i > 0$, $i = 1, 2, 3$, and $g_0 = 9.81$ [m/s²].

This is a 3P robot. Gravity acts on q_3



Exercise #4

Consider the planar 3R robot with equal link lengths in Fig. 2 and absolute joint coordinates $q = (q_1, q_2, q_3)$ defined therein. The robot is equipped with three motors producing torques $u = (u_1, u_2, u_3)$ that perform work on the Denavit-Hartenberg angles $\theta = (\theta_1, \theta_2, \theta_3)$, respectively. Provide the expression of the torque vector $u_q = (u_{q1}, u_{q2}, u_{q3})$ that should appear on the right-hand side of the robot dynamic model written in terms of the absolute coordinates q (i.e., in $M(q)\ddot{q} + n(q, \dot{q}) = u_q$), as function of the components of u .

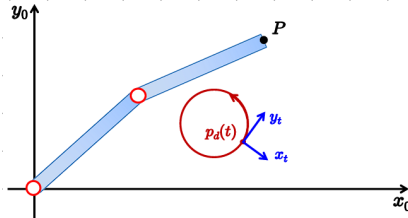


$$\text{We have: } M\ddot{\theta} + n = u_\theta$$

$$q = \mathcal{S}(\theta) = \begin{cases} \theta_1 \\ \theta_1 + \theta_2 \\ \theta_1 + \theta_2 + \theta_3 \end{cases} \Rightarrow J = \frac{\partial \mathcal{S}}{\partial \theta} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\Rightarrow u_q = J^{-T} u_\theta = J^{-T} u = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{cases} u_1 - u_2 \\ u_2 - u_3 \\ u_3 \end{cases}$$

With reference to Fig. 3, the end-point P of a planar 2R robot should execute a desired trajectory $p_d(t) \in \mathbb{R}^2$, specified by a circular path in the Cartesian plane with a time-varying desired tangential velocity. Motion occurs on a horizontal plane and the circle should be traced counterclockwise. An initial position and/or velocity error between the robot end-point and the desired trajectory is present at $t = 0$. Later on, external disturbances may also affect occasionally the execution of the desired motion task. Design a torque control law for the robot such that the trajectory tracking error dynamics is exponentially stable, linear, and decoupled along the normal and tangential directions to the path. The error behaviors in these two directions (represented, respectively, by the x_e and y_e axes of a time-varying frame that moves with the desired task) should be critically damped, with a dominant reaction time to position errors in the normal direction which is about five times faster than in the tangential one. Assume that no kinematic singularities are encountered (as suggested also by the placement of the circle in Fig. 3).



We have the model $M\ddot{q} + c + \cancel{g} = \tau$, we express it in the cartesian space

$$M_c \ddot{\zeta} + C_c = J^T \tau = F$$

$\downarrow$ control force

First of all we apply FBL:

$$F = M_c \ddot{a} + C_c \Rightarrow$$

$$\tau = J^T M_c \ddot{a} + J^T C_c \quad \text{resulting in } \ddot{\zeta} = \ddot{a}. \quad \text{This is expressed in } RF_0$$

$${}^0\ddot{\zeta} = \ddot{a} \quad \text{we want to express it in } RF_t$$

$\Rightarrow RF_t$ is a frame rotated around z of a time varying angle:

$${}^0R_t = \begin{pmatrix} c_\alpha & s_\alpha \\ -s_\alpha & c_\alpha \end{pmatrix} \quad \text{where } \alpha(t) \in [0, 2\pi] \quad \text{so}$$

$${}^0\ddot{\zeta} = \ddot{a} \Rightarrow {}^t\ddot{\zeta} = {}^0R_t^T {}^0\ddot{\zeta} = {}^0R_t^T \ddot{a} = \begin{cases} \ddot{a}_1 c_\alpha - \ddot{a}_2 s_\alpha = \begin{cases} b_1 \\ b_2 \end{cases} = b \\ \ddot{a}_1 s_\alpha + \ddot{a}_2 c_\alpha = \end{cases}$$

$\swarrow$ normal dir
 $\nwarrow$ tang.

So we control b that is translated in commands to a .

$$\Rightarrow {}^t\ddot{\zeta} = b \quad \text{we set } b = K_p(\zeta_d - {}^t\zeta) + K_D(\dot{\zeta}_d - {}^t\dot{\zeta}) + \ddot{\zeta}_d$$

PD law + FeedForward. We have

$${}^t\ddot{\zeta} = K_p e + K_D \dot{e} + \ddot{\zeta}_d \Rightarrow \ddot{e} = \ddot{\zeta}_d - {}^t\ddot{\zeta} \Rightarrow {}^t\ddot{\zeta} = \ddot{\zeta}_d - \ddot{e}$$

$$\ddot{e} + K_p e + K_D \dot{e} = 0 \Rightarrow \lim_{t \rightarrow \infty} e = 0$$

It should be critically damped so: $K_{D2}^2 = 4K_{p2}$

The normal dir. should be recovered five times faster.

$$\text{time const} = \lambda_i = \sqrt{\frac{K_{p_i}}{2}} \quad \text{and we want } \lambda_1 = 5\lambda_2$$

$$\Rightarrow \text{We set } K_{p1} = \alpha \quad K_{D1} = \sqrt{4\alpha}, \quad \alpha > 0$$

$$\lambda_1 = \sqrt{\frac{\alpha}{2}} \Rightarrow \lambda_2 = 5\sqrt{\frac{\alpha}{2}} = \sqrt{\frac{K_{p2}}{2}} \Rightarrow K_{p2} = 25\alpha \Rightarrow K_{D2} = \sqrt{100\alpha}$$